

Program Elective-II**Course Code- 151607 Cloud Computing Network****3 0 0 3****Unit-1.0: Introduction to Cloud Computing**

Definition and characteristics of cloud computing, Cloud service models (IaaS, PaaS, SaaS), Deployment models (Public, Private, Hybrid, Multi-cloud), Cloud computing architecture, Virtualization in cloud computing, Hypervisors (VMware, KVM, Xen).

Unit-2.0: Cloud Networking Basics

Cloud network architecture, Software-Defined Networking (SDN), Network Functions Virtualization (NFV), Virtual Private Cloud (VPC), Content Delivery Networks (CDN), Cloud load balancing and traffic management.

Unit-3.0: Cloud Storage and Data Management

Cloud storage models (Object, Block, File storage), Distributed file systems (Google File System, Hadoop HDFS), Cloud database services (NoSQL, NewSQL), Data consistency and replication, Cloud backup and disaster recovery.

Unit-4.0: Cloud Security and Compliance

Cloud security challenges, Identity and Access Management (IAM), Encryption in cloud, Secure cloud networking (VPN, TLS, SSL), Compliance standards (ISO 27001, GDPR, HIPAA), Zero Trust security in cloud environments.

Unit-5.0: Cloud Deployment and Orchestration

Containerization (Docker, Kubernetes), Serverless computing, Cloud automation (Terraform, Ansible), Multi-cloud and hybrid cloud management, Microservices architecture in cloud, Edge computing and IoT integration.

Unit-6.0: Emerging Trends and Case Studies

AI and Machine Learning in cloud computing, Blockchain for cloud security, Quantum computing and cloud, Case studies on real-world cloud deployments (AWS, Azure, Google Cloud), Future trends in cloud networking.

Text/Reference:

1. Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, Mastering Cloud Computing, McGraw Hill.
2. Thomas Erl, Ricardo Puttini, Zaigham Mahmood, Cloud Computing: Concepts, Technology & Architecture, Pearson.
3. Anthony T. Velte, Toby J. Velte, Robert Elsenpeter, Cloud Computing: A Practical Approach, McGraw Hill.
4. Arshdeep Bahga & Vijay Madisetti, Cloud Computing: A Hands-on Approach, Universities Press.
5. Kai Hwang, Geoffrey Fox, Jack Dongarra, Distributed and Cloud Computing, Elsevier.
6. George Reese, Cloud Application Architectures: Building Applications and Infrastructure in the Cloud, O'Reilly.
7. Benedict Blundell, Docker and Kubernetes for Cloud Networking, Packt Publishing.